

CompE565, Spring 2026

HW 1

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List of Figures

Figure 1: Original image (1200x1600) and RGB Bands

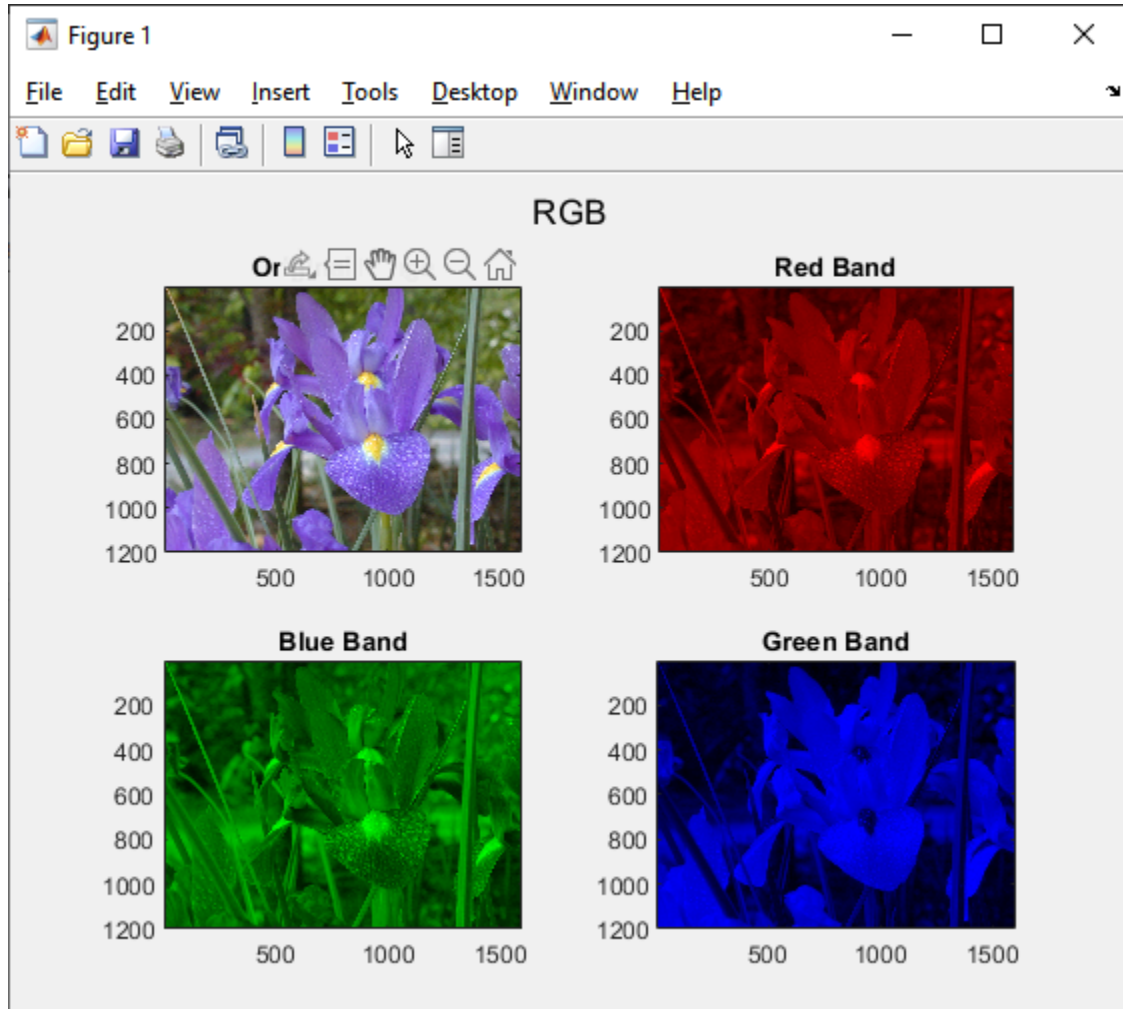


Figure 2: YCbCr image (1200x1600) and YCbCr Bands

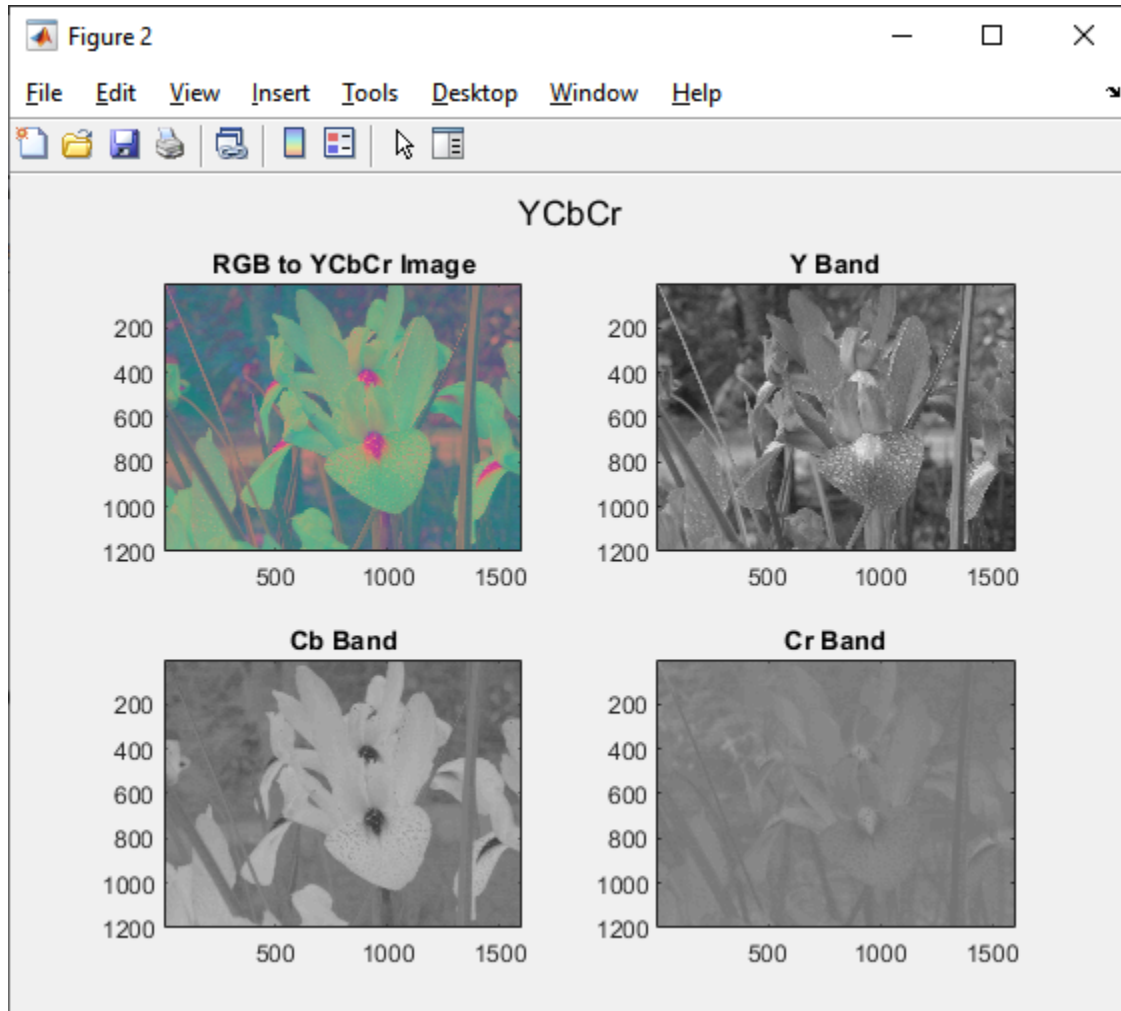


Figure 3: Subsampled CbCr Bands

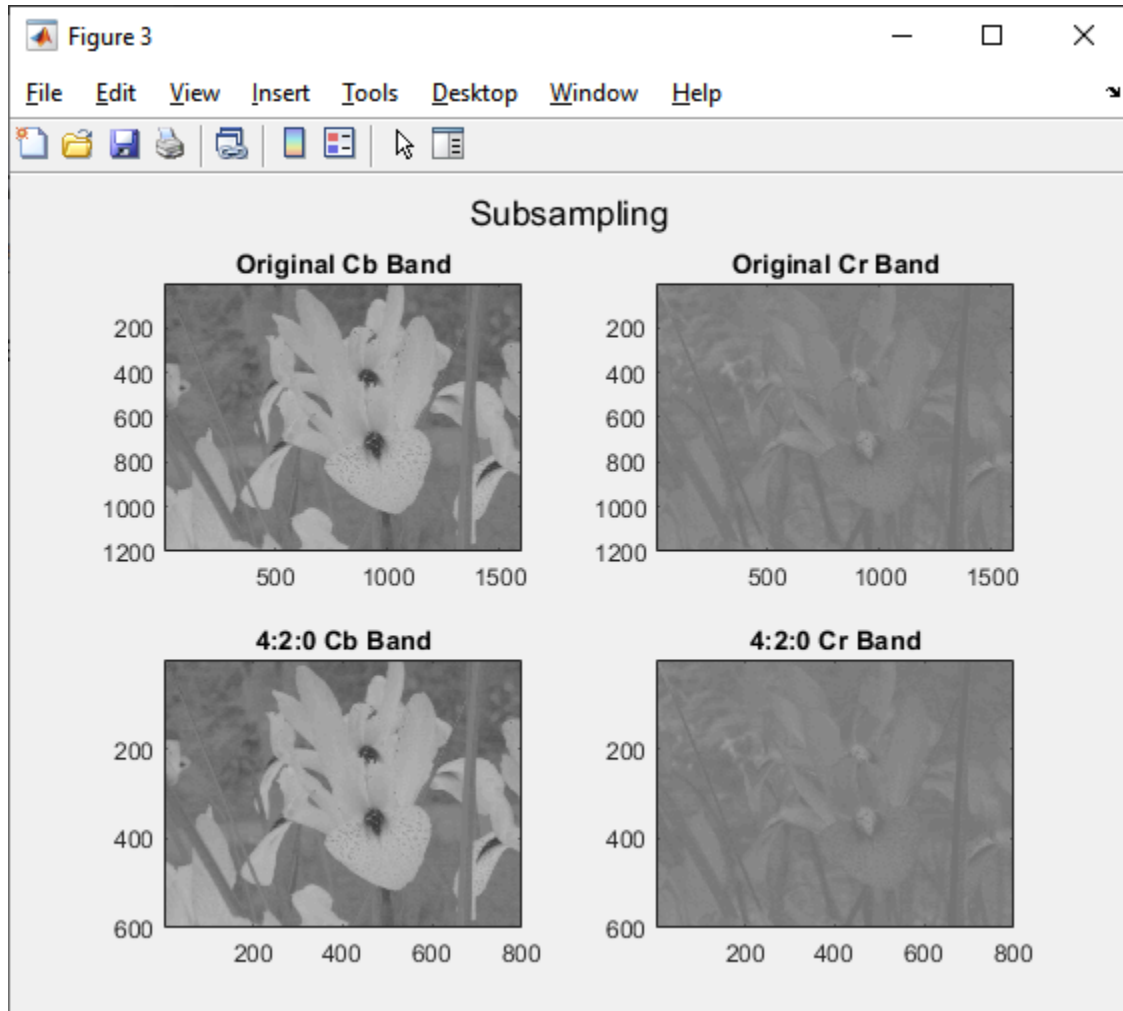


Figure 4: Linear Interpolated CbCr Bands

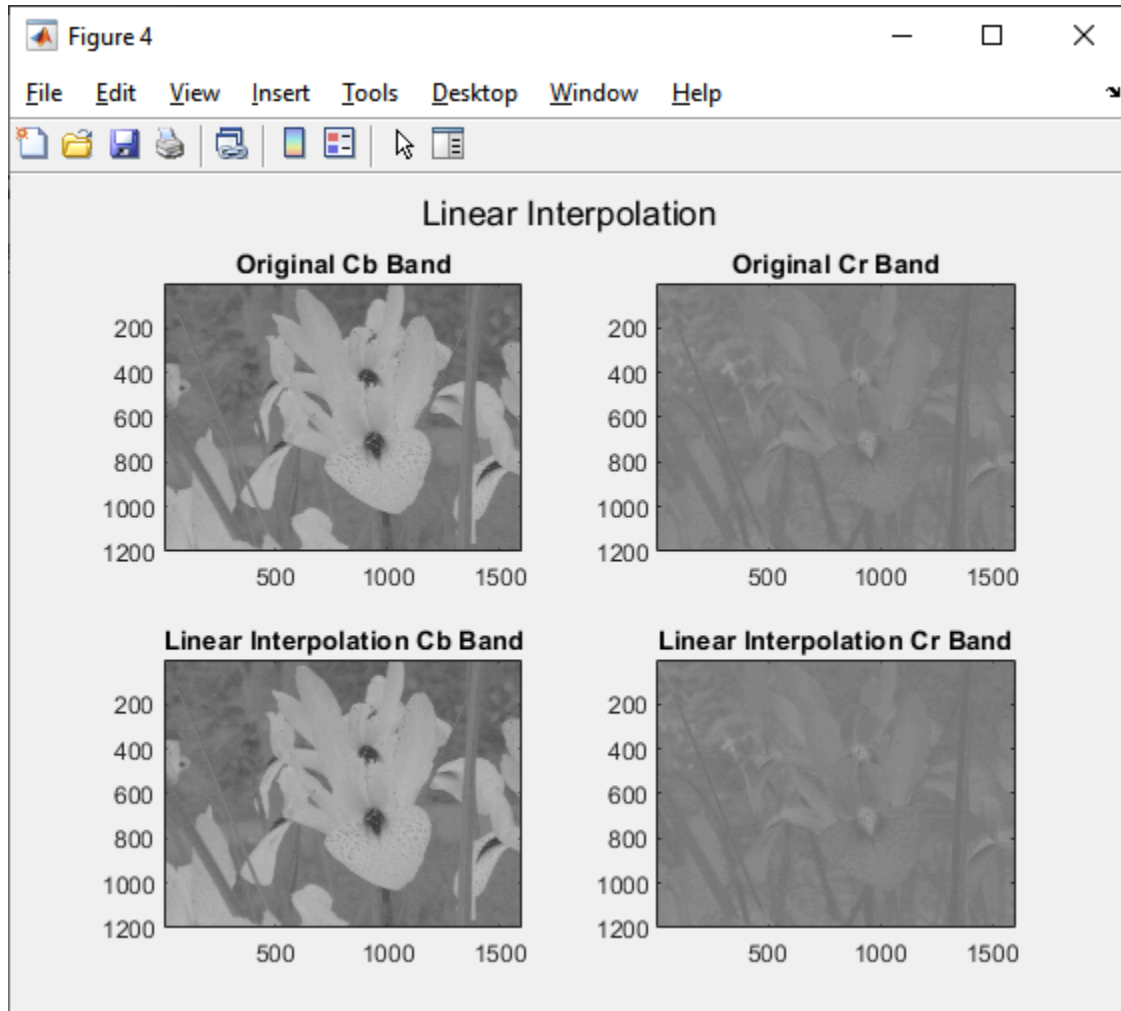


Figure 5: Row/Column Replicated Bands

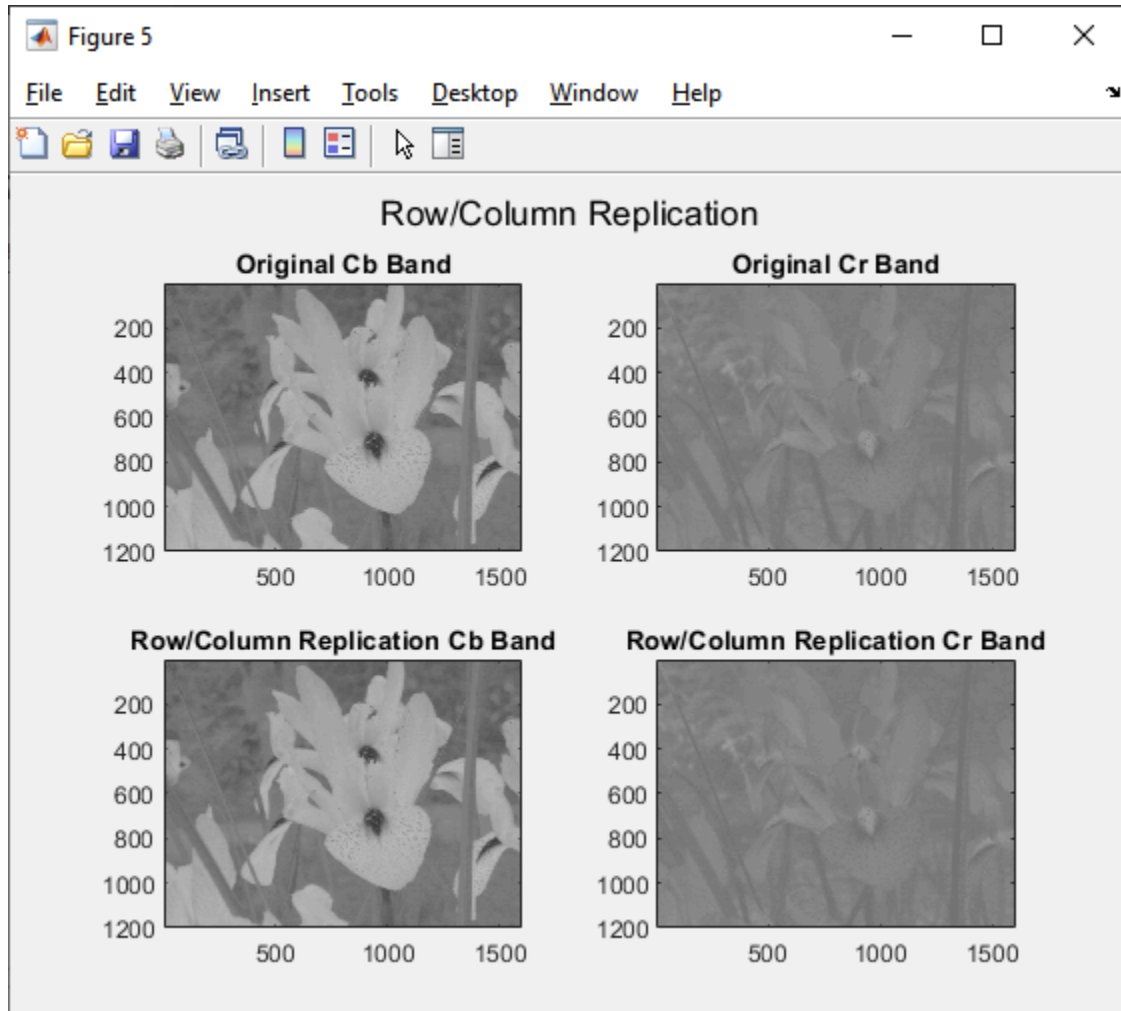
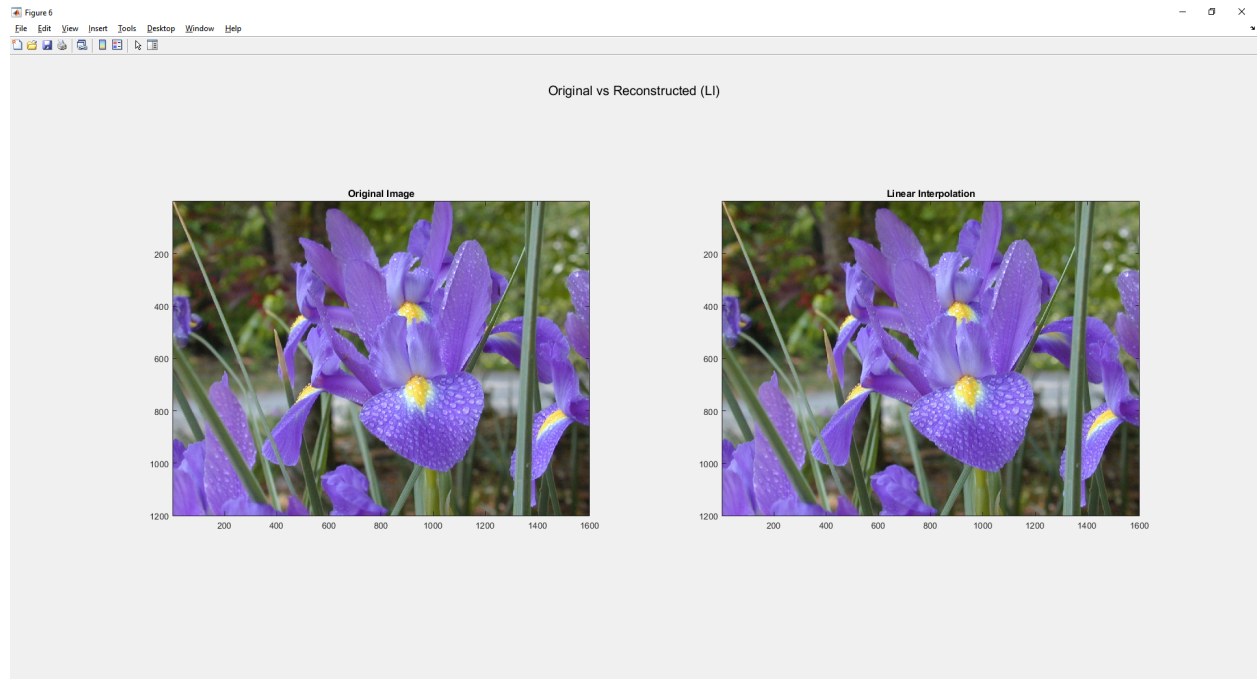


Figure 6: Original vs Reconstructed Image



List of Tables

Table 1: Output of MSE and PSNR

MSE	1.344372
PSNR	46.845608

Introduction

Storing images and videos at full resolution requires substantial storage capacity because high-resolution content contains millions of pixels. This large volume of data increases storage costs and slows down transmission. Compression addresses this issue by removing redundant information in visual data, enabling more efficient storage and faster transfer speeds while preserving acceptable visual quality.

This work lies within the field of digital image processing, with an emphasis on compression techniques. In particular, it examines color space conversion, chroma subsampling, and upsampling methods.

The main objective of this assignment is to explore how images can be compressed without significantly degrading their visual appearance. The 4:2:0 chroma subsampling format is applied, and the reconstructed image is compared to the original to evaluate quality. This topic is important because effective compression makes it possible to store large datasets more efficiently and transmit media over bandwidth-limited channels.

Compression plays a vital role in modern digital media systems. Chroma subsampling is widely implemented in formats such as JPEG for images and MPEG for video. The YCbCr color space is commonly used in these standards, with 4:2:0 being one of the most prevalent subsampling schemes.

The following section of this report outlines the methodology used to complete the assignment. The subsequent section presents the results, and the report concludes with a summary of findings and a list of references.

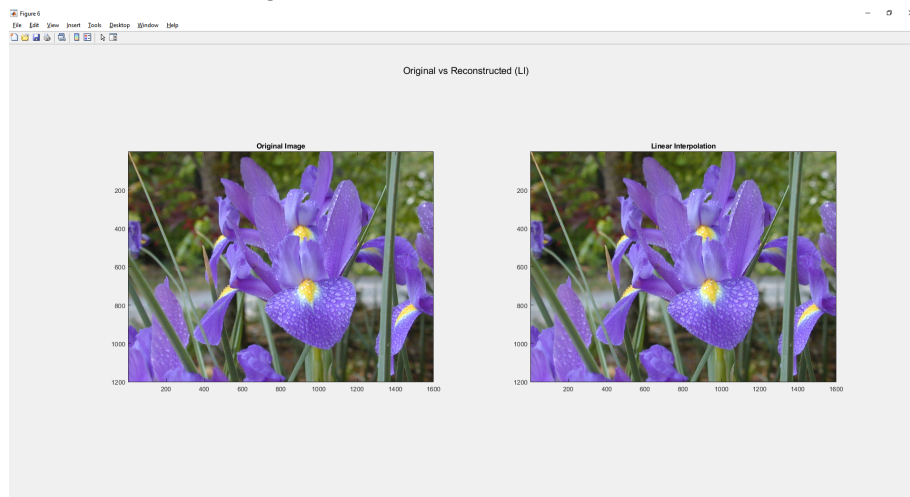
Procedural Section

The program takes in any input image (the program can handle different dimensions), these are the steps taken to subsample and upsample a given image (Iris.jpg):

1. Input
 - a. Reads Iris.jpg
2. Color Space Conversion (Figure 1 and 2)
 - a. Extract RGB bands
 - b. Convert to YCbCr
 - c. Extract YCbCr bands
3. Subsampling (Figure 3)
 - a. 4:2:0 subsample the Cr and Cb bands
4. Upsampling (Figure 4 and 5)
 - a. Linear interpolation
 - b. Row/Column replication
5. Reconstruction
 - a. Convert upsampled YCbCr back to RGB
6. Calculate Statistics (Table 1)
7. Output (Figure 6)
 - a. Figures
 - b. MSE and PSNR

Results

The results throughout this are shown above in the list of figures and tables. The final results of MSE, PSNR, and reconstructed images are also shown below.



MSE	1.344372
PSNR	46.845608

Conclusion

This project developed a simple image compression workflow based on YCbCr color space conversion and 4:2:0 chroma subsampling. The findings indicate that lowering the chrominance resolution provides a 2:1 compression ratio while preserving visual fidelity as shown by the low MSE and high PSNR measurements. Both interpolation and replication techniques for upsampling were able to reconstruct the image with only minor visual differences. These results highlight the practical value of chroma subsampling in real-world image and video compression systems.

References

N/A